

NAICHENG JIANG

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RESEARCH INTERESTS

3D and spatial intelligence; multimodal learning, vision-language models, and NLP; reliable and feedback-driven AI; Clinical world models.

EDUCATION

KTH Royal Institute of Technology, Stockholm, Sweden Aug 2025 - Jun 2027 (expected)
MSc in Information and Network Engineering, School of Electrical Engineering and Computer Science
Current weighted grade average: **4.72/5.00** (*self-calculated from completed graded courses*)

Shanghai Jiao Tong University, Shanghai, China Aug 2022 - Jun 2026
BSc in Electrical and Computer Engineering, UM-SJTU Joint Institute (Global College)
GPA: **3.57/4.00**

RESEARCH EXPERIENCE

VecSetX - 3D Anatomical Representation and Completion from Partial Observations Feb 2026 - Nov 2026(expected)
Research Collaborator

- Built and validated key components of a unified **30-class 3D anatomical representation** across **four heterogeneous datasets**, including spatial-orientation harmonization, label-space alignment, and partial-supervision handling.
- Designed **cross-dataset and leak-safe completion experiments**, including controlled source comparisons, supervision-weight ablations, full-resolution evaluation, and paired statistical analysis.
- Analyzed **conditional generative completion** and identified **weak use of contextual information** as a key bottleneck, motivating more reliable completion from incomplete 3D observations.

Pulse - Reliable 3D Pulmonary-Tree Analysis and Structural Quality Assessment Jul 2026 - Present
Research Collaborator

- Led reproducible **3D airway failure characterization** on a **50-case ATM22 diagnostic cohort**, focusing on connectivity, component structure, confidence, and structural errors.
- Showed that naive largest-component filtering can improve aggregate metrics while removing valid, high-confidence airway structures, revealing a **structural reliability issue not captured by Dice alone**.
- **Planned next stage**: develop an **inference-time structural evaluator** that provides localized feedback for candidate repair and re-evaluation; **GT-free assessment** and **natural-language feedback** are longer-term directions.

MANUSCRIPT IN PREPARATION

VecSetX — **Co-first author**; manuscript in preparation for submission to **CVPR 2027**, deadline Nov 2026.

HONORS & AWARDS

- **Outstanding Graduate**, Shanghai Jiao Tong University, 2026
- Selected for the **GIFT Future Scholars Program**, Shanghai Jiao Tong University, 2024
- **Yu Liming Scholarship**, UM-SJTU Joint Institute, 2024

TECHNICAL SKILLS

Programming: Python, MATLAB, C++
Machine Learning & Vision: PyTorch, OpenCV, NumPy/SciPy
Research Computing: Linux, Git, LaTeX, Slurm, LUMI/HPC, CUDA
Languages: Chinese (native), English (professional working proficiency)